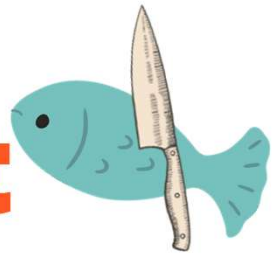
Reasoning About CUT Feasibility from Physical Properties



Team Affordance
Alex Clark, Hyejin Park, Liangliang Zhang, Xiye Fan





1

Average Score: 3.99/5

Danielle Villa, 2026/10/8

Project Overview

- Ontology explains how the **physical properties**, **object parts**, and **relationships between parts** supports or prevents the physical actions or capabilities of a given object
- A researcher supplies structured descriptions of candidates—their known properties, relevant parts, and part relationships
- Evaluation Rule:
    - **Feasible** – configuration satisfies all requirements of action
    - **Infeasible** – every evaluated configuration of action is blocked by known evidence
    - **Insufficient Information** – No conclusion
- The system **does not** recognize objects, estimate properties from images, or derive capabilities from conventional functions
- Scoped to only the **CUT** action – extend to more actions if time permits
- Assumes an ordinary human actor in an everyday kitchen setting



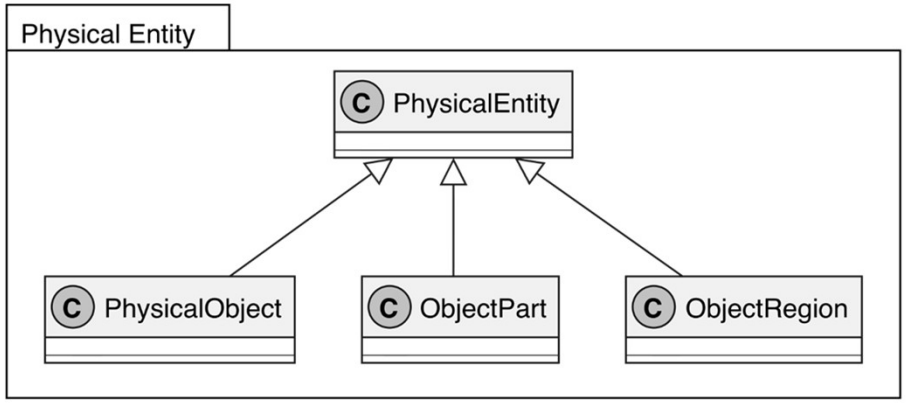
Competency Questions

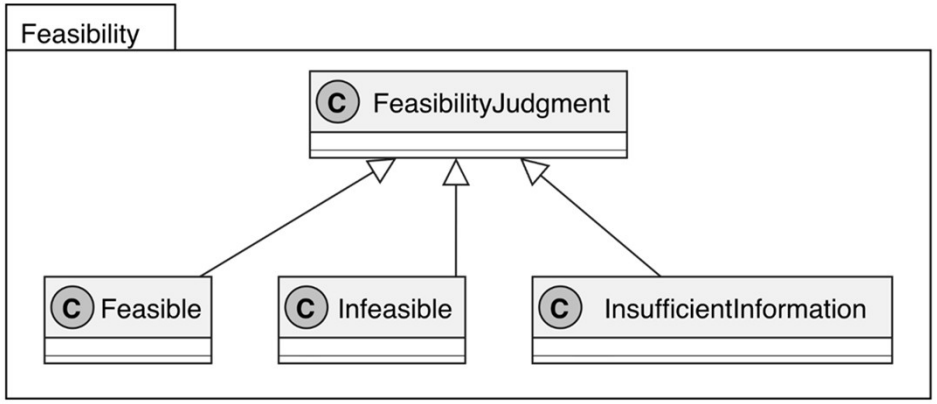
- **CQ1.1 —** ***Can a rigid plastic card be used by an ordinary human actor to cut a carrot, and why?***
- **CQ1.2 —** ***Can a typical kitchen sponge be used to cut a carrot, and why or why not?***
- **CQ2.1 —** ***If a conventional cutting tool is unavailable, which of the following available objects could satisfy the represented physical requirements for cutting a carrot: a rigid plastic card, a rolling pin, or a kitchen sponge?***
- **CQ2.2 —** ***Why can a rigid plastic card be returned as a candidate for cutting a carrot while a kitchen sponge is excluded?***
- ***CQ3.1 — Can a knife and a pair of scissors both support CUT despite having different relevant part structures and part relationships?***

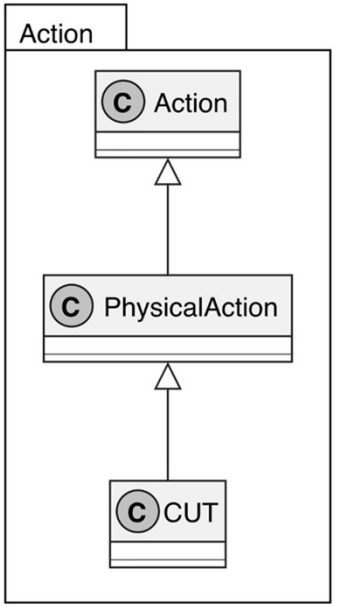


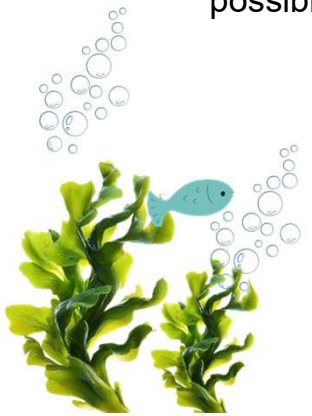
Conceptual Model — Classes

- **Physical Entity:** represents the physical things involved in the task, including `PhysicalObject`, `ObjectPart`, and `ObjectRegion`.
- **Action:** represents the activity being evaluated, with `CUT` modeled as a specific `PhysicalAction`.
- **Feasibility:** represents the outcome of the evaluation, with `Feasible`, `Infeasible`, and `InsufficientInformation` as possible judgments.



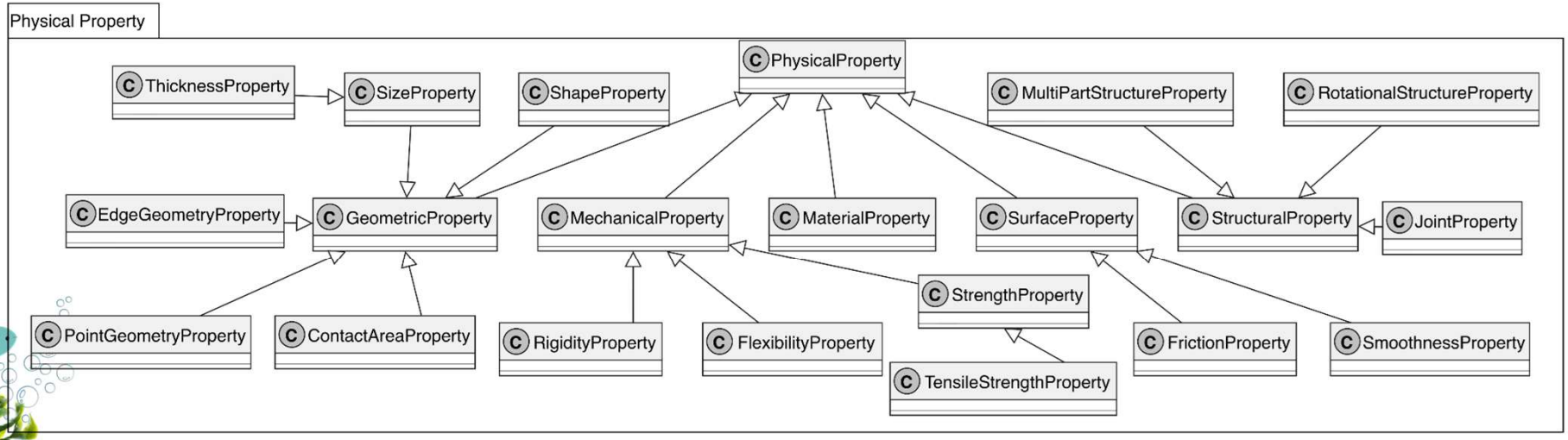






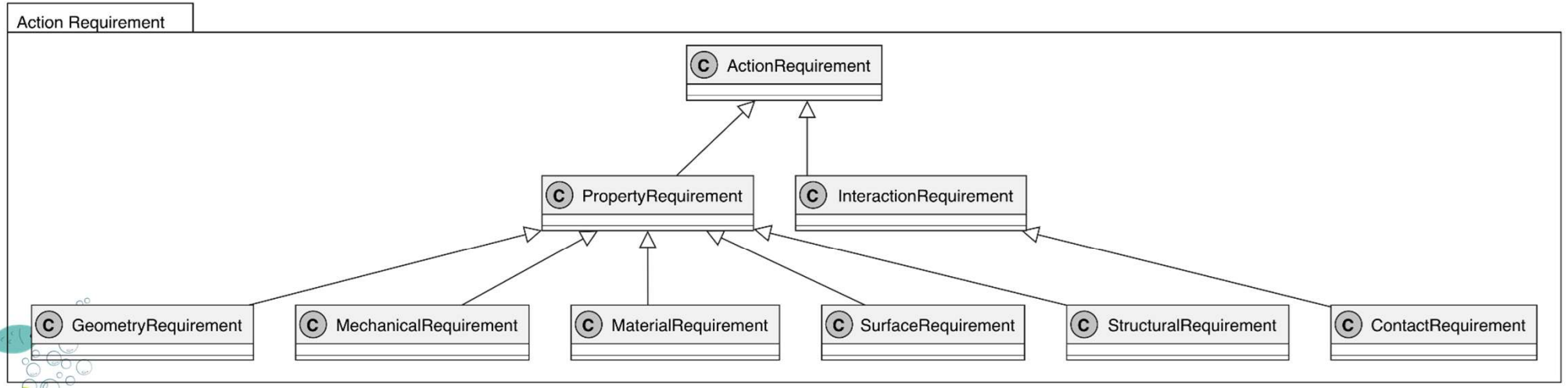
Conceptual Model — Classes

- **Physical Property:** represents the physical characteristics used for feasibility reasoning.
  - **Geometric properties:** capture shape, size, thickness, edge geometry, point geometry, and contact area.
  - **Mechanical properties:** capture rigidity, flexibility, strength, and tensile strength.
  - **Surface properties:** capture friction and smoothness.
  - **Structural properties:** capture joints, rotational structures, and multi-part structures.
  - **MaterialProperty** remains a separate physical-property category for material-related characteristics.



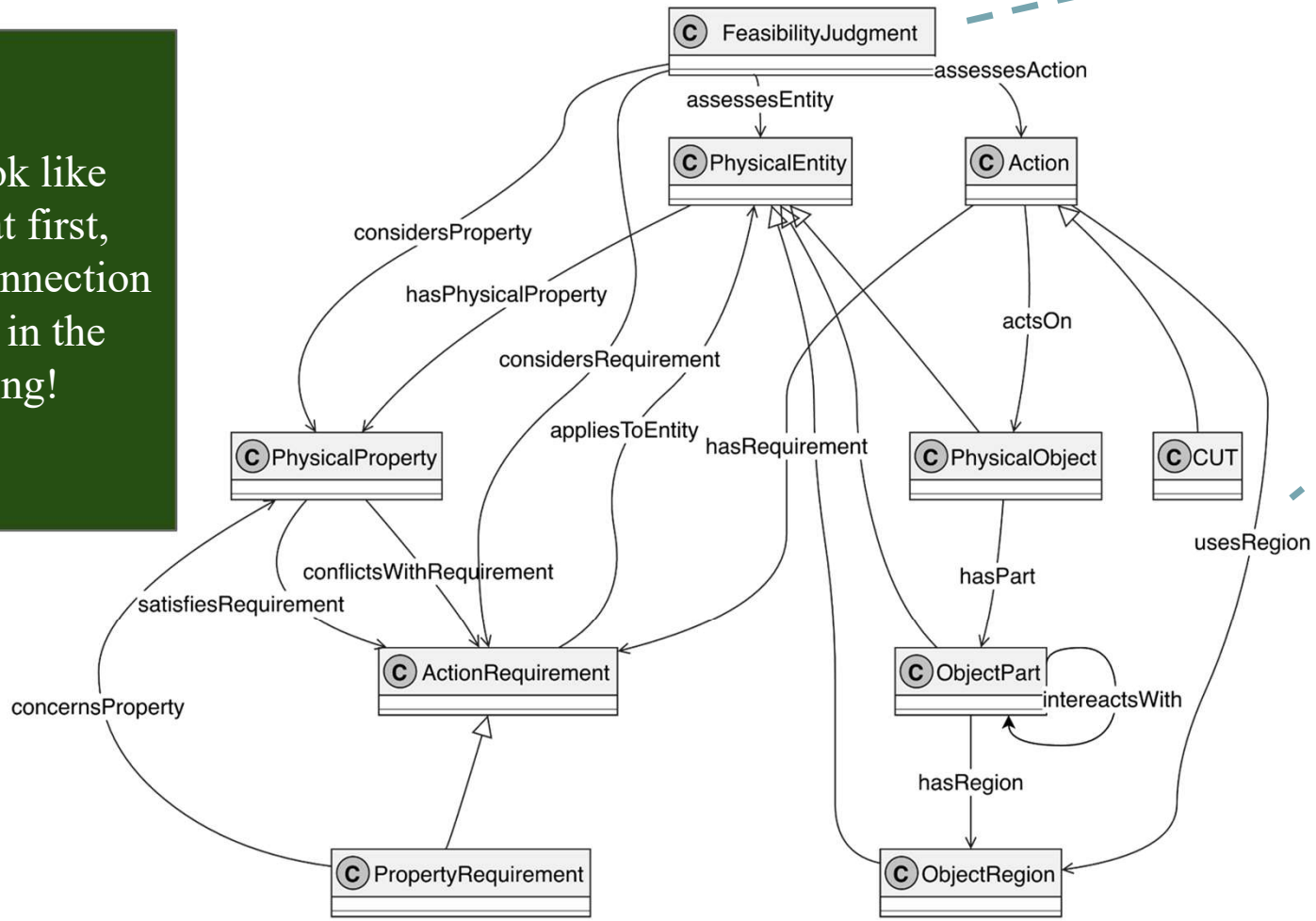
Conceptual Model — Classes

- **ActionRequirement** captures the conditions needed for an action to be feasible.
  - **PropertyRequirement** covers geometry, mechanical, material, surface, and structural conditions.
    - 6 requirements
  - **InteractionRequirement** captures how the action physically interacts with the object.
    - **ContactRequirement** specializes interaction requirements related to physical contact.

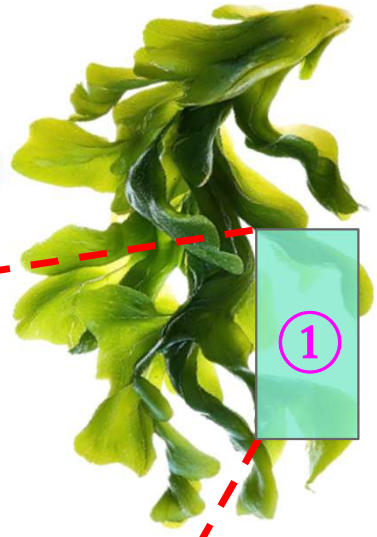


Our SEAWEED Relation Diagram

It may look like seaweed at first, but every connection has a role in the reasoning!



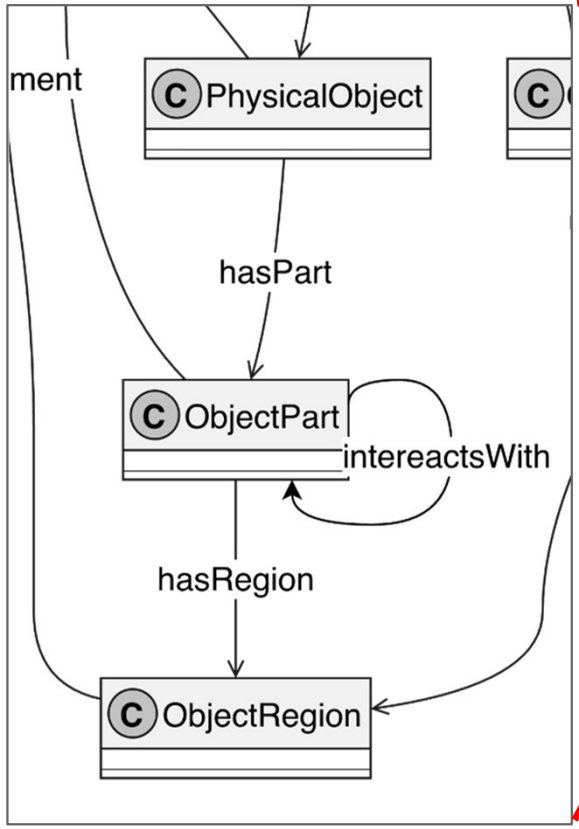
Idea 1: Find the Relevant Physical Structure



Let's break the seaweed into three ideas.

1. Identify the relevant physical structure

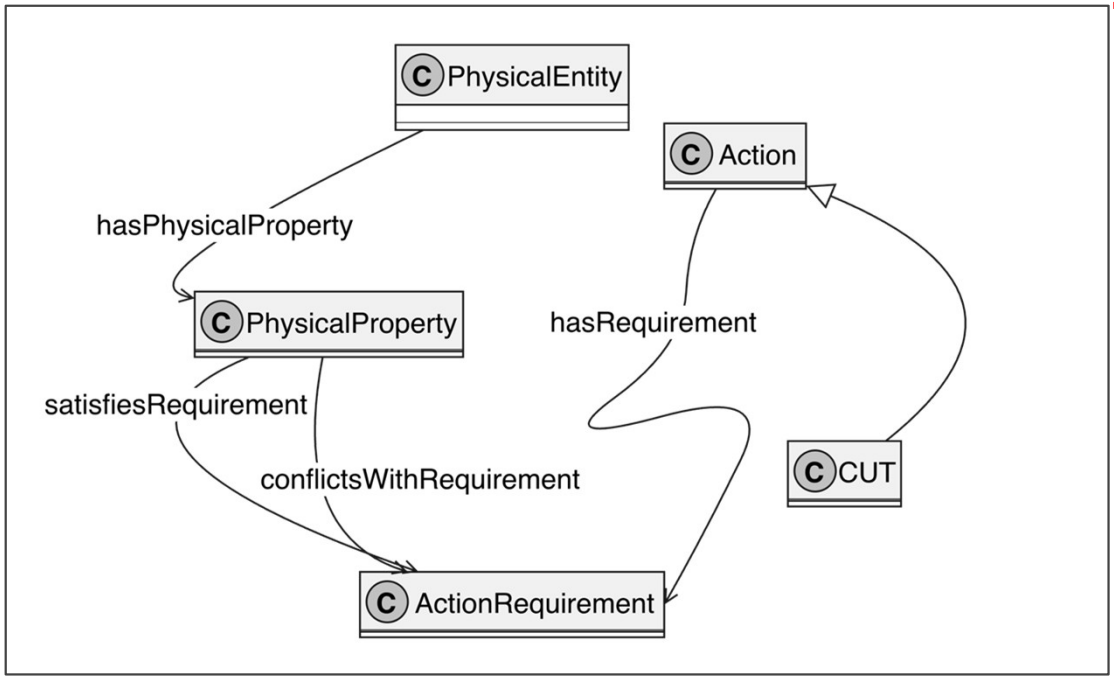
- An object may contain multiple parts and regions.
- CUT may depend on a specific edge, surface, or interacting part.
- Therefore, the ontology reasons about relevant parts rather than the object as a whole.

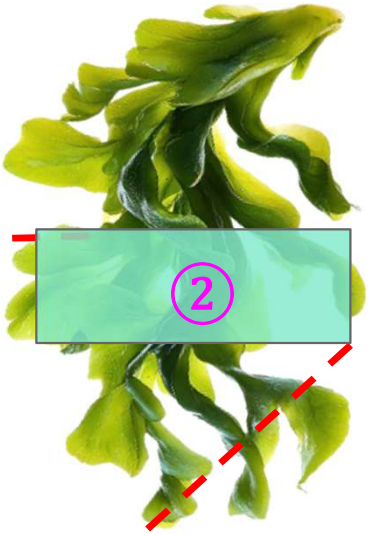






Idea 2: How the comparison works





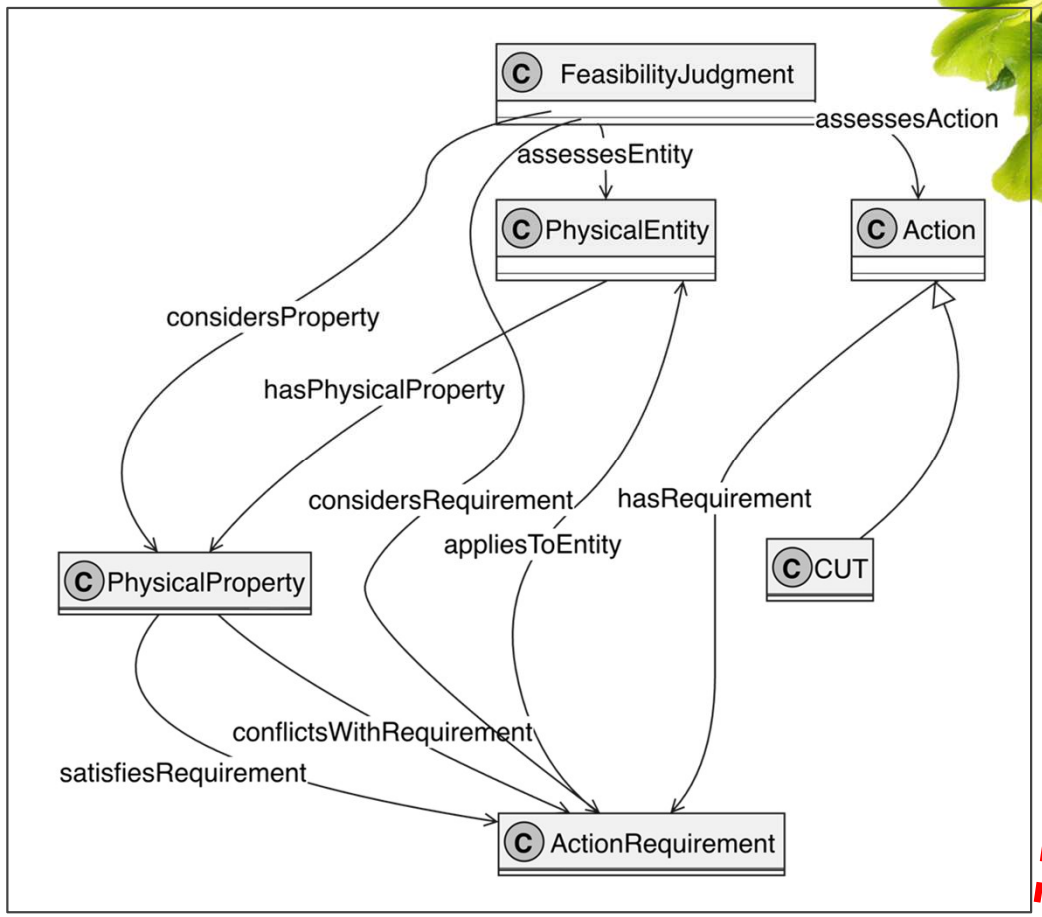
2. Compare physical properties with CUT requirements

- CUT is associated with physical requirements.
- Candidate properties are evaluated against these requirements.
- Properties may satisfy or conflict with a requirement.
- Missing information is not automatically treated as a conflict

Idea 3: How the final judgment is built

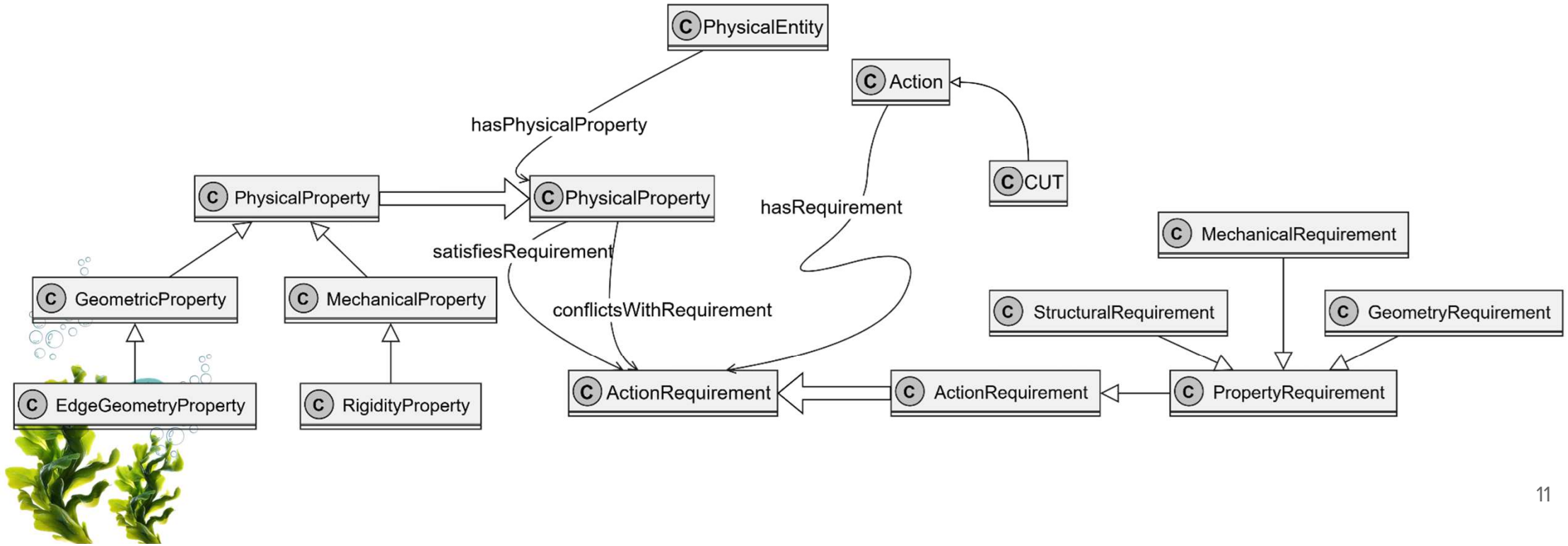
3. Produce an explainable feasibility judgment

- The judgment considers the candidate physical description
- It records the relevant properties and requirements.
- Supporting and conflicting evidence can be traced.
- This supports feasible, infeasible, or insufficient-information results.

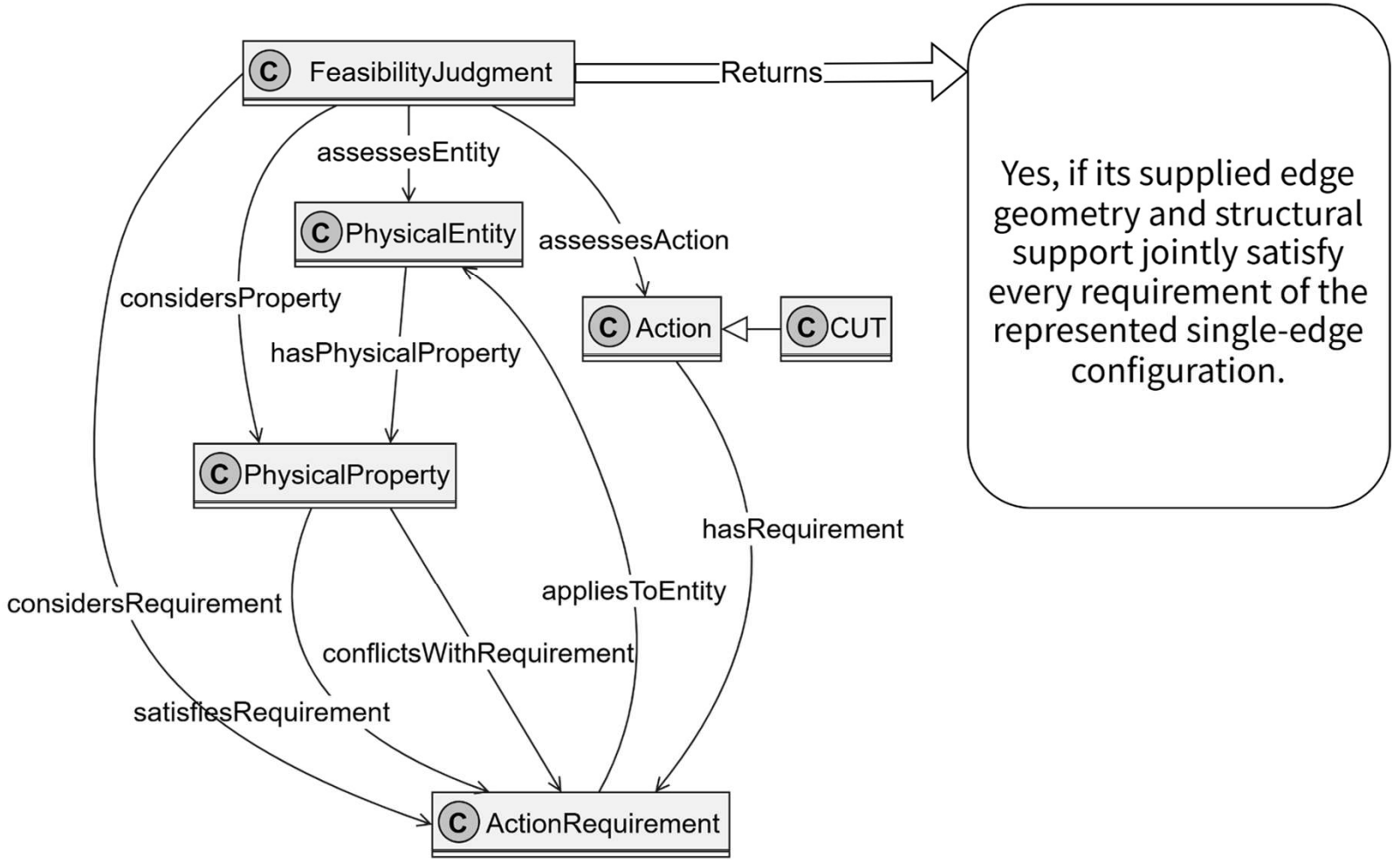


How the Model Addresses the CQs

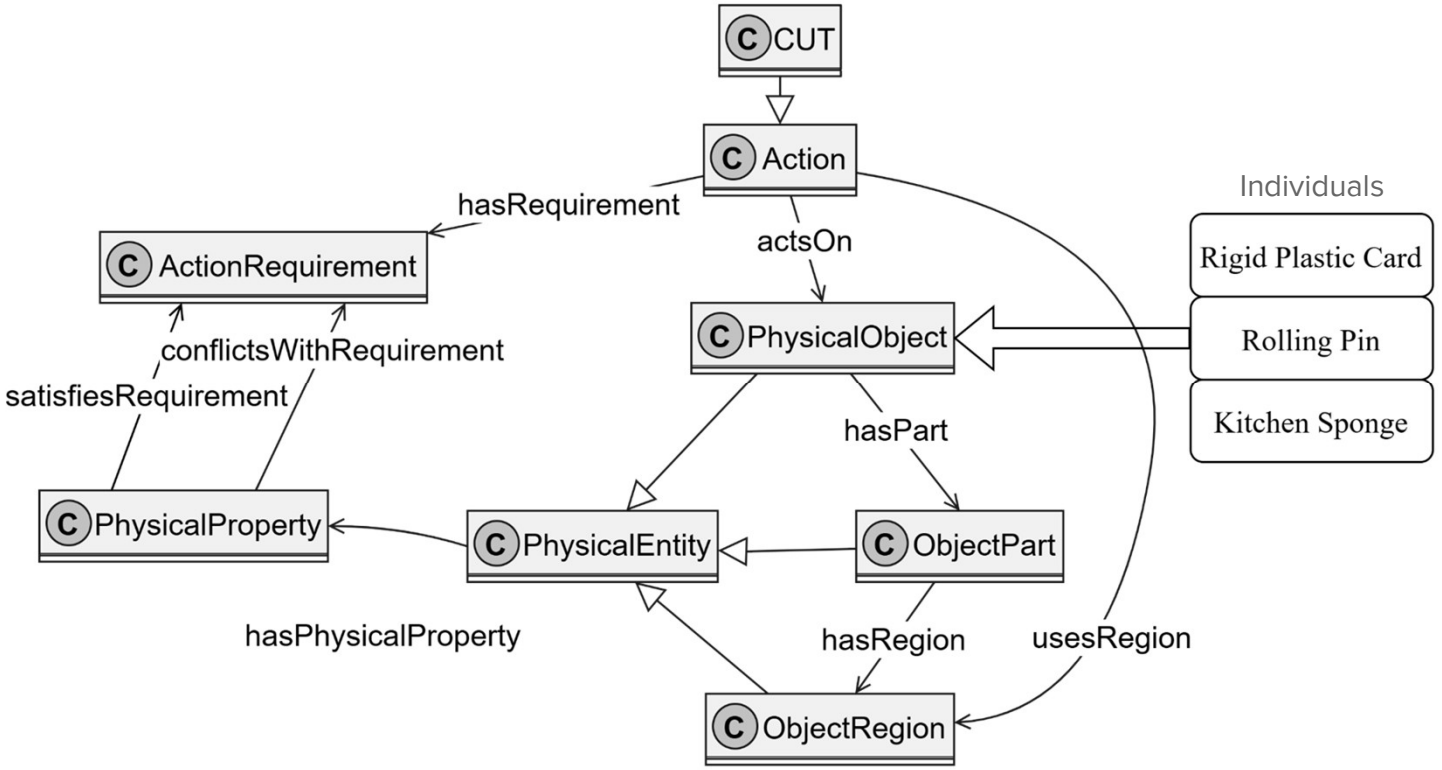
Example CQ 1.1: ***Can a rigid plastic card be used by an ordinary human actor to cut a carrot, and why?***



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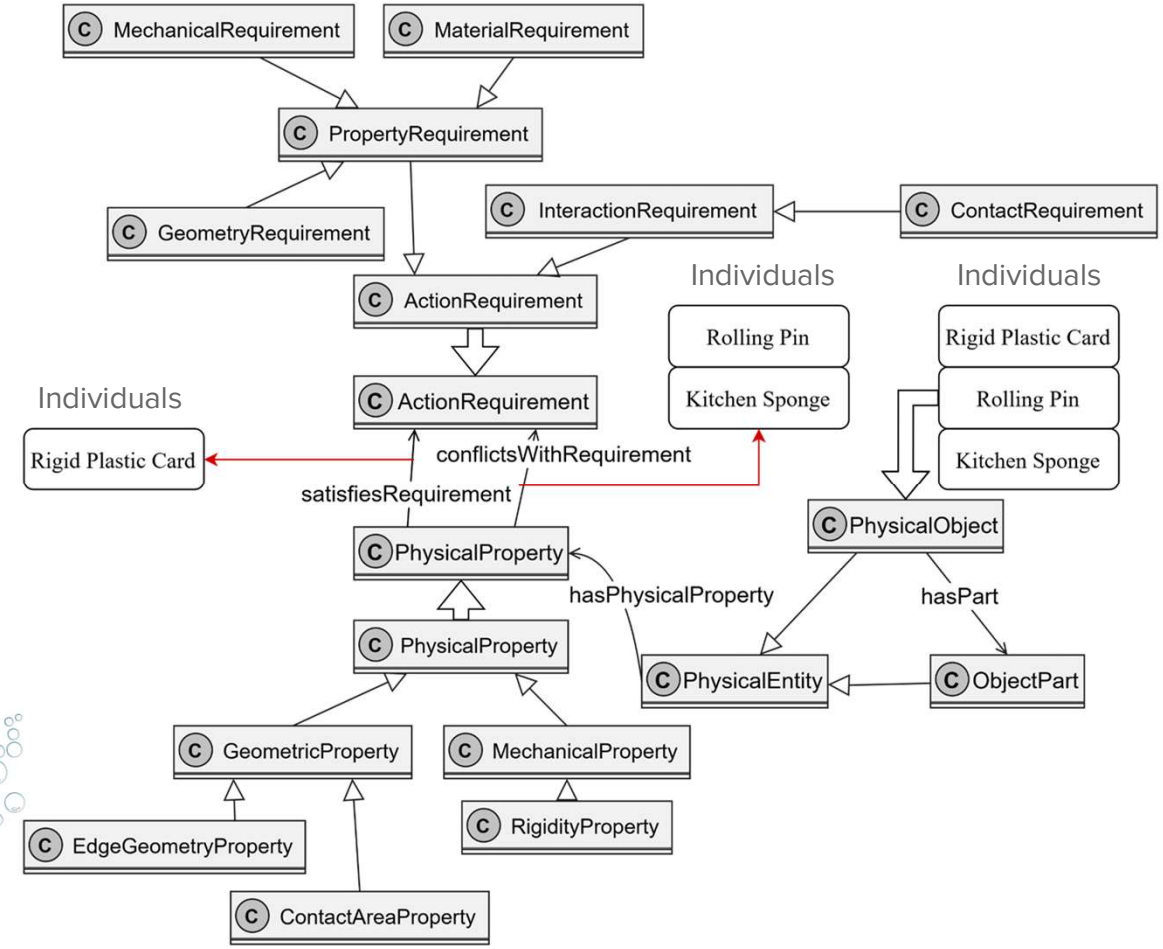


Example CQ 2.1: ***If a conventional cutting tool is unavailable, which of the following available objects could satisfy the represented physical requirements for cutting a carrot: a rigid plastic card, a rolling pin, or a kitchen sponge?***



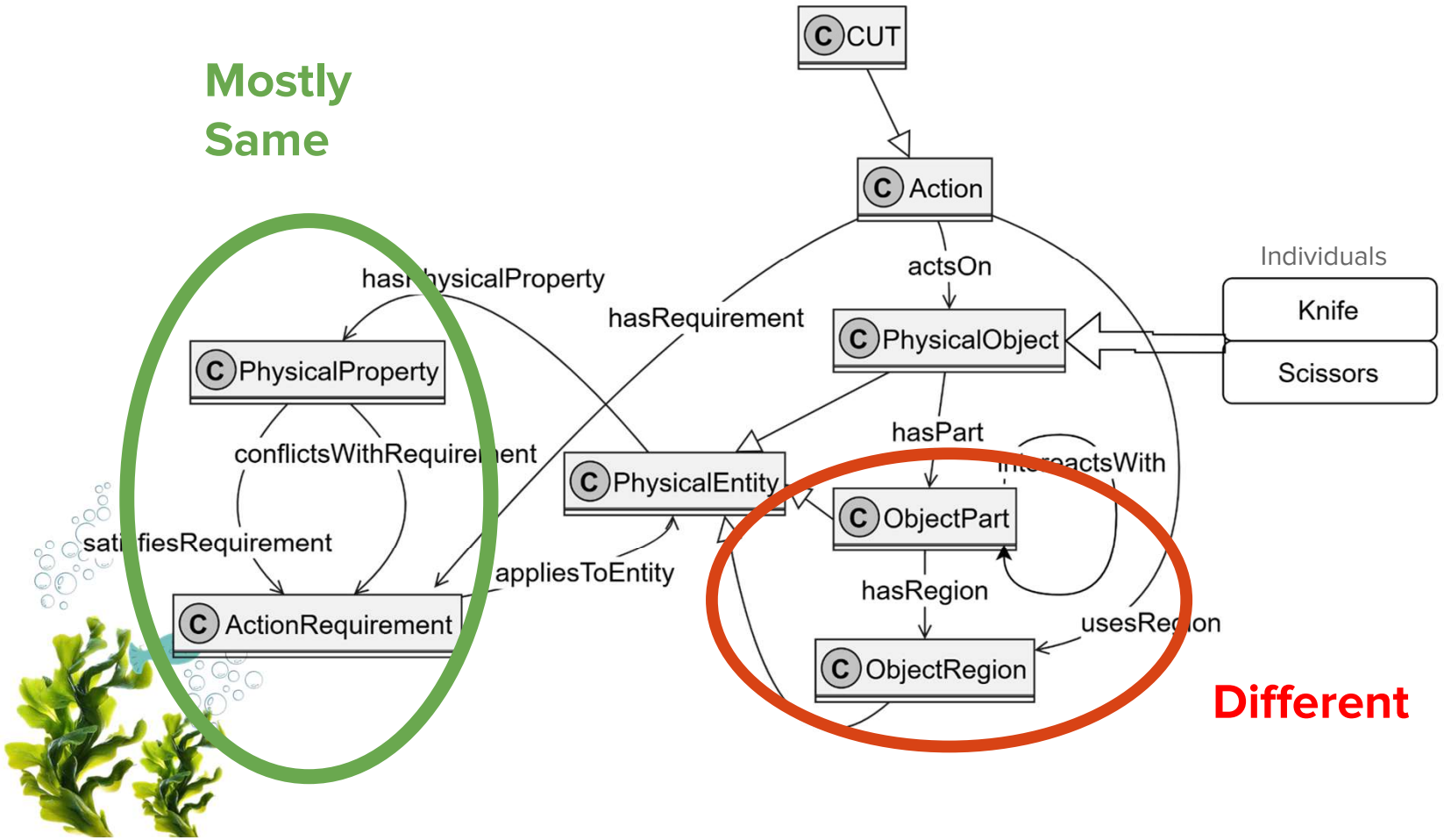


Example CQ 2.1: ***If a conventional cutting tool is unavailable, which of the following available objects could satisfy the represented physical requirements for cutting a carrot: a rigid plastic card, a rolling pin, or a kitchen sponge?***



Under the represented conditions, the rigid plastic card is returned as a candidate because its edge satisfies the modeled geometry and rigidity requirements. The rolling pin and kitchen sponge are excluded because their represented properties conflict with different cutting requirements.

Example CQ 3.1: ***Can a knife and a pair of scissors both support CUT despite having different relevant part structures and part relationships?***



Yes. Both may support CUT through different combinations of relevant physical properties and part relationships. A knife may rely on a single relevant edge with suitable geometry and structural support, while scissors may rely on two relevant parts whose properties and interaction jointly satisfy the applicable CUT requirements.

How We Used GenAI for the Conceptual Model

1. We Provided

- Revised use case
- Curated terminology
- Modeling constraints

2. GenAI (ChatGPT) Helped Us Explore

- Suggested candidate class and relationship structures
- Generated a draft diagram from text/code
- Provided ideas for layout and organization

3. We Reviewed & Rebuilt

- Checked whether the suggested relationships matched our model
- Used the draft mainly as a visual reference
- Redrew the conceptual model ourselves in draw.io

